# TRACKING CLIMATE-RELATED DEVELOPMENT FINANCE, 2015-2023

## MAJOR TRENDS, KEY ACTORS, AND REMAINING GAPS

Kyuin LEE

2015 marked a pivotal year in climate diplomacy, when countries made a near-universal commitment to curb global warming. A decade on, the world is on the brink of overshooting the 1.5°C limit. Yet progress toward a timely, effective, and coordinated response has proven difficult, as climate policies have constantly shifted with the political pendulum. Against this backdrop, this report examines the evolution of climate-related development finance over 2015–2023, focusing on the scale of climate finance flows, the major providers and recipients, and the gaps that remain.


## SUMMARY

- Climate issues appear to have gained renewed momentum since 2021. Total climate finance commitments have steadily increased, while funding allocated to projects with an explicit climate focus has followed a U-shaped trajectory. Although the scope of this study excludes the Trump 2.0 era, the latest observations point to a sustained positive momentum across 2015-2023.
- Major providers of climate-related development finance are:
  - Multilateral development banks (MDBs): the lending institutions of the World Bank and the Asian Development Bank
  - Bilateral donors: Japan, Germany, the EU, France, and the US
- Philanthropic institutions and non-MDB multilateral institutions play a modest role in the climate finance landscape.
- Climate finance is mostly concessional, but an increasing share of climate finance is offered on non-concessional terms. In Latin America & the Caribbean and Europe, recipients already receive more non-concessional than concessional finance. If climate finance further aggravates the severe debt burdens of developing countries, it will serve as a roadblock to escaping the low- and middle-income trap.
- Major recipients of climate finance vary depending on the metric:
  - Total commitment: South-East Asian countries such as India, Bangladesh, the Philippines, and Indonesia were the largest recipients. It was noteworthy that China, which has recently emerged as an important donor within the

development cooperation landscape, was also a major recipient — as far as climate finance was concerned.
    - Commitment per capita: SIDS received the largest amounts of climate finance. Among non-SIDS, several countries in Latin America & the Caribbean (Costa Rica, Panama) and Eastern Europe (Georgia, Montenegro, Serbia, Albania) received the highest per capita allocations.
- Climate finance is moderately responsive to the *economic* vulnerabilities and risks of developing countries, but allocation decisions should more closely align with their *environmental* needs and realities.

# GLOBAL TRENDS

Throughout 2015-2023, **climate-related development finance[1] has followed a clear upward trajectory**, reaching a record $145.11 billion in 2023 (Figure 1). The volume of climate finance plateaued during the first year of COVID-19 but soon picked up, showing a sizeable increase since.

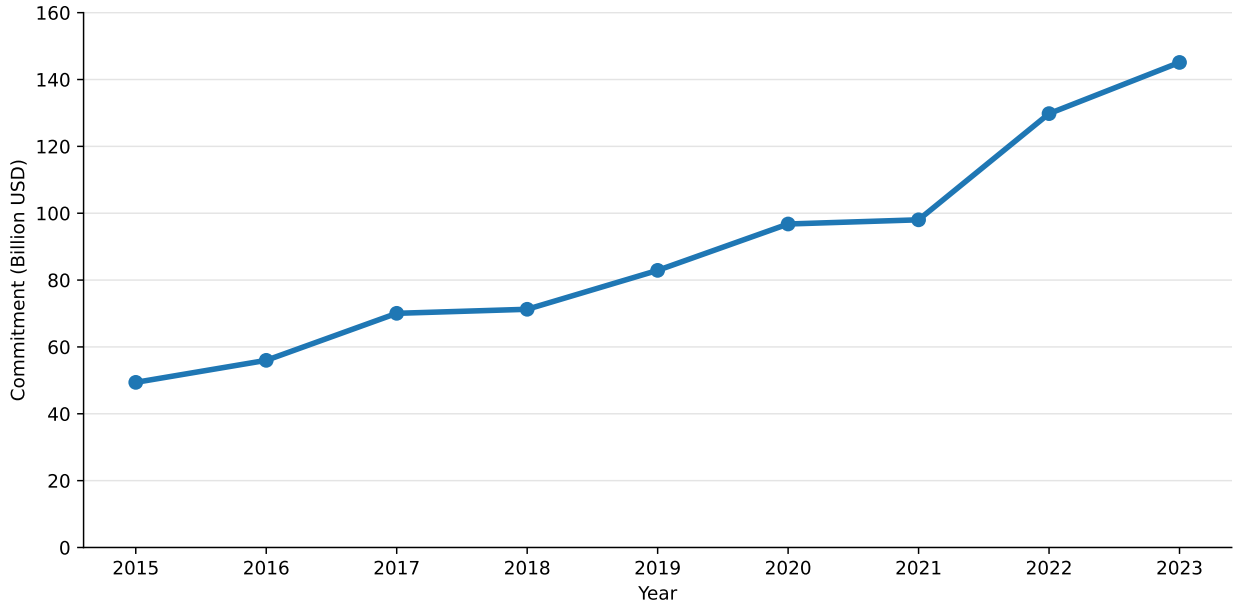


Figure 1: Annual Climate Finance Allocations

[1] Commitments are financial pledges that reflect donor intentions, whereas disbursements are the actual payments made, representing the realization of those commitments. The figures reported herein refer to climate finance commitments (adjusted for inflation; constant 2023 prices), as they are better suited to understanding donors' priorities.

In terms of region, ***Asia* was the largest recipient of climate finance** (Figure 2). Lately, however, some of the **flow seems to have been diverted towards *Africa***, the second largest recipient.

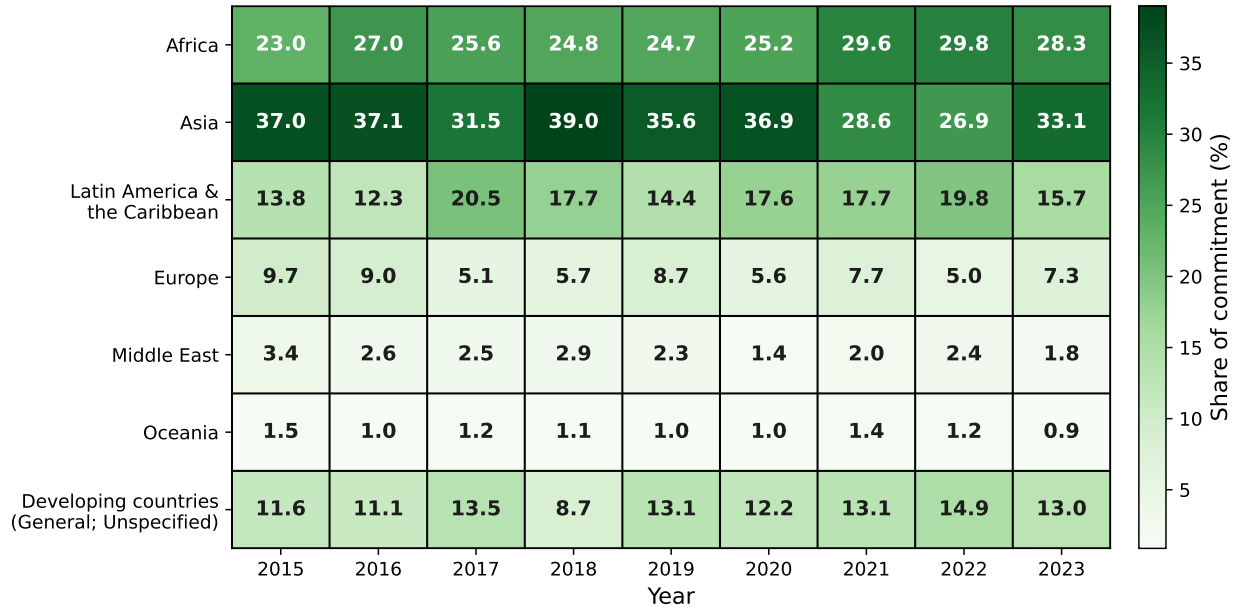


Figure 2: Share of Climate Finance Flows, by Region

On a positive note, **climate issues have regained prominence since 2021**, as evinced by both the increasing share of, and the amount of finance allotted to projects with climate as the principal objective[2] (Figure 3; Figure 4). These shifts coincide with the onset of the Biden administration, whose pro-climate agenda appears to have contributed to a global momentum toward ambitious climate action. Data is not yet available beyond 2023, but whether this positive momentum has persisted into the Trump 2.0 era will be an important point of inquiry for future research.

Amid this renewed emphasis on climate, mitigation-focused projects have consistently received more finance than adaptation-focused projects (Figure 4). In terms of project count, however, the two were mostly comparable throughout 2015-2023 (Figure 3). This suggests that **mitigation projects tend to be more capital-intensive**, often involving larger-scale investments such as grid expansion and power plant construction.

[2] Development projects were assessed based on the extent to which they pursue climate change adaptation and mitigation — whether as principal objectives, secondary objectives, incorporating some climate component, or not targeting climate objectives.

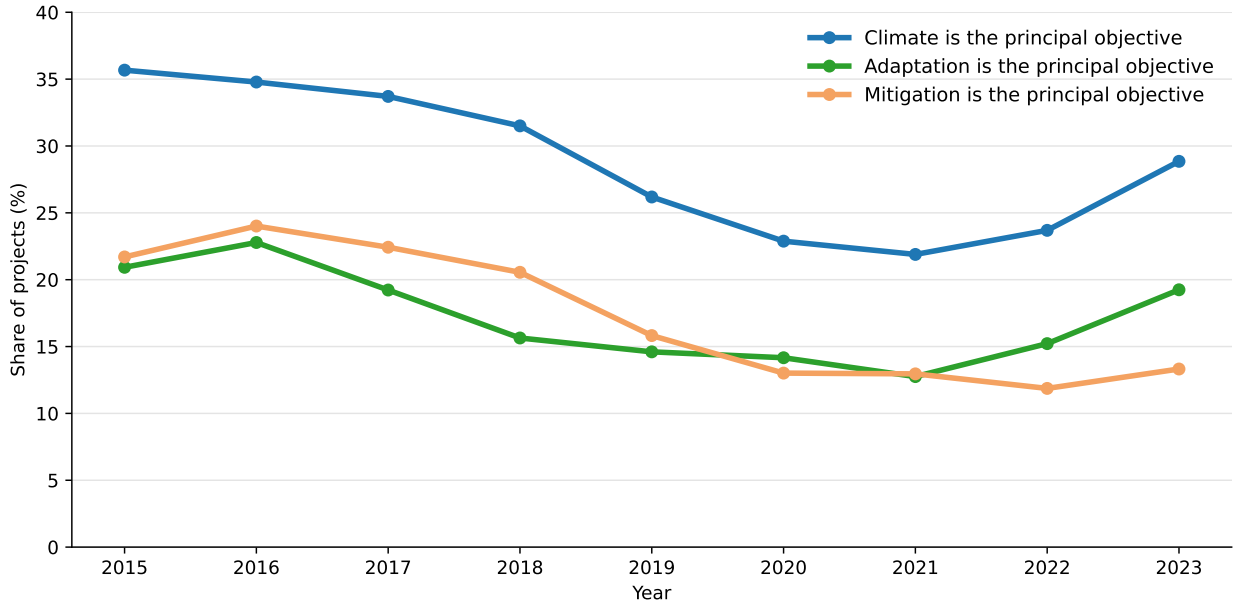


Figure 3: Share of Development Projects that Identify Climate as the Principal Objective

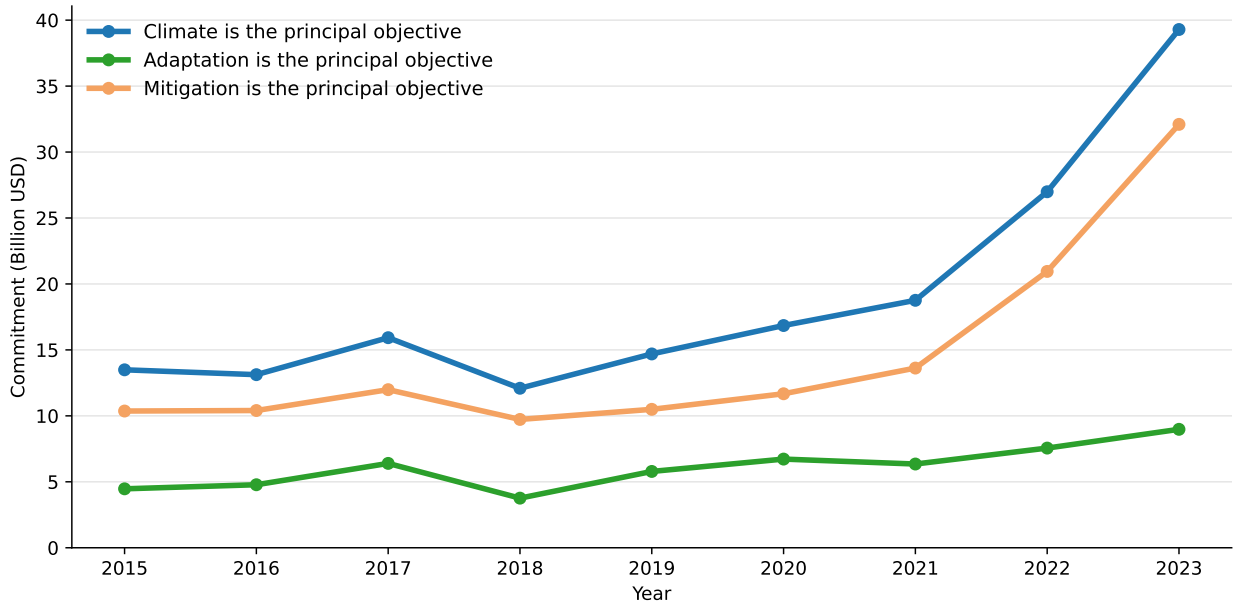


Figure 4: Amount of Finance Allocated to Projects with Climate as the Principal Objective

# MULTILATERAL PROVIDERS OF CLIMATE FINANCE

Given that multilateral development banks (MDBs) are essentially 'financial' institutions that provide loans and technical assistance to support countries in need, **MDBs played a prominent role in the climate finance landscape, relative to non-MDB multilaterals** (Figure 5; Figure 6). In fact, MDBs emerged as the largest source of climate finance since 2021, surpassing bilateral donors (governments)[3]. Meanwhile, the latter's share decreased from 59% in 2015 to 45% in 2023.

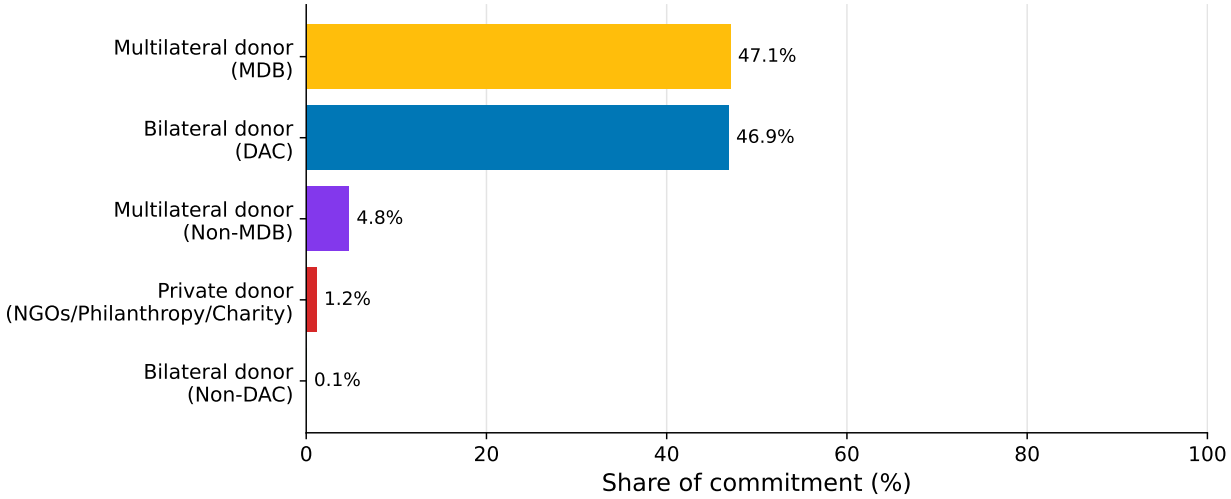


Figure 5: Sources of Climate Finance, Total (2015-2023)

Among multilateral providers, three arms of the World Bank (WB)[4] — namely *International Bank for Reconstruction and Development* (IBRD), *International Development Association* (IDA), and *International Finance Corporation* (IFC) — ranked among the five largest sources of climate finance (Table 1). Also notable was the *Asian Development Bank* (ADB), which was the third-largest source of climate finance despite its narrow geographic scope (Asia and Oceania).

[3] Bilateral providers consist of two types: (1) Members of the OECD Development Assistance Committee (DAC) and (2) Non-DAC members. Often referred to as traditional donors, members of the OECD DAC are primarily high-income countries that follow a common set of principles and reporting standards for development aid. There are 33 DAC members: the *EU*, *Australia*, *Austria*, *Belgium*, *Canada*, *Czechia*, *Denmark*, *Estonia*, *Finland*, *France*, *Germany*, *Greece*, *Hungary*, *Iceland*, *Ireland*, *Italy*, *Japan*, *Latvia*, *Lithuania*, *Luxembourg*, *the Netherlands*, *New Zealand*, *Norway*, *Poland*, *Portugal*, *the Slovak Republic*, *Slovenia*, *South Korea*, *Spain*, *Sweden*, *Switzerland*, *the UK*, and *the US*.

[4] The three institutions of the WB play different but complementary roles: IBRD lends to *middle-income* countries and creditworthy low-income countries, IDA provides concessional finance to the world's *poorest countries*, and IFC supports the *private sector* in developing countries.

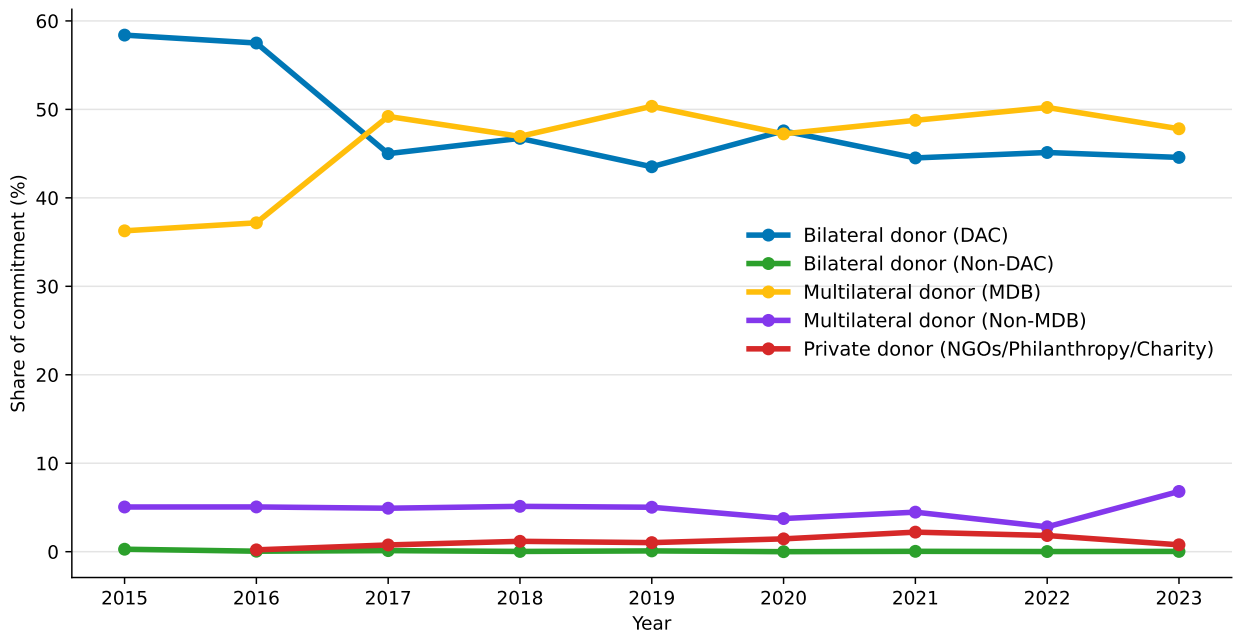


Figure 6: Sources of Climate Finance, Time Series

Table 1: Top 10 Multilateral Donors

| Rank | Provider | Commitment |
|---|---|---|
| 1 | International Bank for Reconstruction and Development | $76.20B |
| 2 | International Development Association | $75.19B |
| 3 | Asian Development Bank | $42.65B |
| 4 | European Bank for Reconstruction and Development | $28.06B |
| 5 | International Finance Corporation | $27.88B |
| 6 | European Investment Bank (EIB) | $27.42B |
| 7 | Inter-American Development Bank | $25.13B |
| 8 | Development Bank of Latin America | $19.14B |
| 9 | Green Climate Fund | $14.47B |
| 10 | Asian Infrastructure Investment Bank | $13.59B |

# BILATERAL PROVIDERS OF CLIMATE FINANCE

**Among bilateral providers[5], *Japan* was the largest contributor of climate finance, followed by *Germany*, the *EU*, *France*, and the *US*** (Table 2). While *Japan*'s climate finance is predominantly focused on Asia (Figure 11), *France* and the *EU* direct a substantial share to Africa (Figure 13, Figure 14). In contrast, the *US* adopts a multi-regional approach, with much of its finance unearmarked at the country-level and instead directed to developing countries more broadly (Figure 15).

Table 2: Top 10 Bilateral Donors

| Rank | Provider | Commitment |
|---|---|---|
| 1 | Japan | $92.67B |
| 2 | Germany | $76.24B |
| 3 | EU Institutions (Excluding EIB) | $49.79B |
| 4 | France | $49.60B |
| 5 | United States | $22.83B |
| 6 | United Kingdom | $15.03B |
| 7 | Netherlands | $11.70B |
| 8 | Norway | $9.47B |
| 9 | Korea | $8.88B |
| 10 | Canada | $6.07B |

As for sectoral priorities, *Japan* had a predominant focus on *Transport*, which accounted for more than half of its climate finance commitments (Table 3). By contrast, the *EU* exhibited a diverse portfolio, with significant allocations to *Agriculture* and *Energy* (Table 5). *Germany* and *France* put greater emphasis on *Energy* as well as *Water Supply & Sanitation*, albeit to a lesser extent (Table 4; Table 6).

Across these major bilateral providers (except for the US), *multi-sector* consistently ranked among the top five sectors (Table 3; Table 4; Table 5; Table 6). This suggests that **climate-related development finance is often structured to address interconnected challenges across multiple sectors.**

As with regional allocation, the *US* was less specific in its sectoral allocation, leaving a significant share of its climate finance unspecified at the sector level (Table 7). Where sectors were identified, however, much of its finance was allocated to *Sexual[6] and Reproductive Health* and *Family Planning*, along with *Agriculture*.

[5] Note that the OECD DAC classifies the EU as a *bilateral* provider, but the European Investment Bank (EIB) — the EU's lending arm — as a *multilateral* provider. Accordingly, all references to EU institutions in this report are treated as bilateral and exclude the EIB, as it falls under a different provider category.

[6] The control of Sexually Transmitted Infections (STIs), including HIV/AIDS.

# CONCESSIONALITY OF CLIMATE FINANCE

**Overall, a majority of climate-related development finance is concessional, but the gap between concessional and non-concessional finance is constantly closing** (Figure 7). A closer look reveals a more troubling trend: In fact, in *Latin America & the Caribbean* and *Europe*, the amount of non-concessional finance has long exceeded that of concessional finance (Figure 16; Figure 17). Furthermore, the fact that MDBs are the largest provider type, and that most of their climate finance (74%) is neither concessional nor primarily aimed at development, adds to existing concerns about **mounting debt burdens of developing countries.**

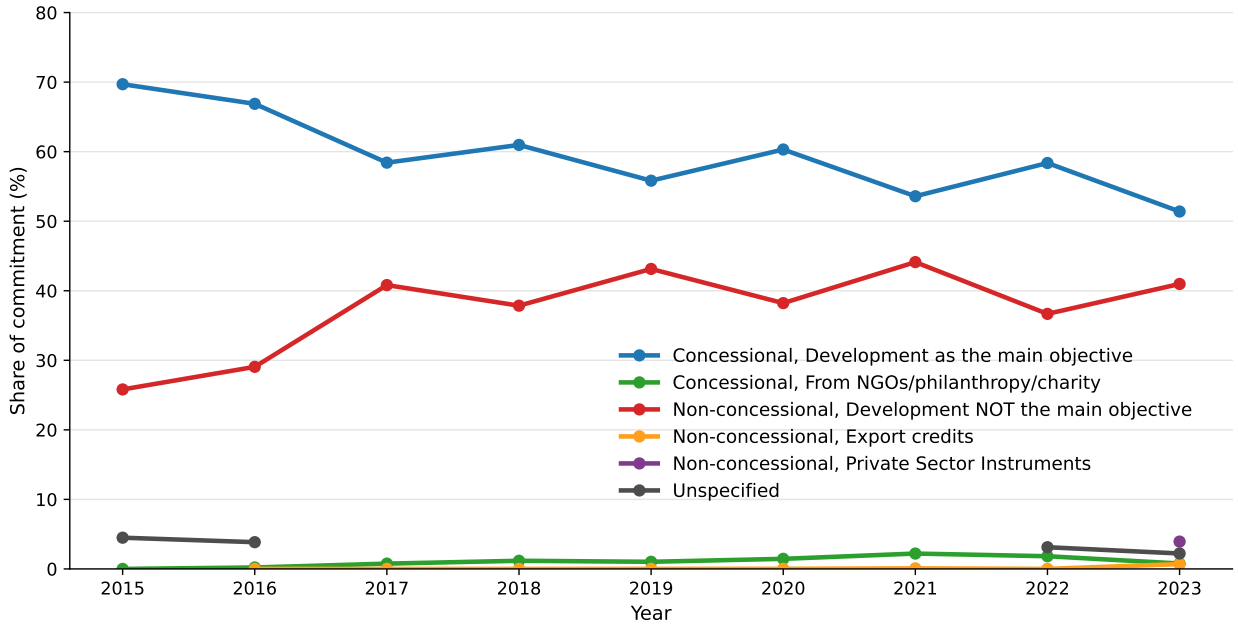


Figure 7: Share of Climate Finance, by Concessionality

Traditional bilateral providers (otherwise known as members of the OECD Development Assistance Committee; DAC) also relied on some non-concessional means, such as export credits[7] and private sector instruments[8]. Meanwhile, despite accounting for only a small

[7] Due to higher risks, exporters are often reluctant to trade with buyers in developing countries. Hence to support domestic exports, donor governments provide export credits to these foreign buyers — that is, government-backed loans, guarantees, or insurance which allow payment to be deferred. Export credits offer mutual benefits: importers gain access to finance, while exporters are protected against commercial (non-payment as a result of bankruptcy or insolvency) and political (non-payment as a result of government intervention or conflict or war) risks.

[8] Private sector instruments (PSI) are financial tools that promote private sector activity in developing countries. They operate via two channels: (1) the provision of *finance to private sector entities* and (2) the *mobilization of private capital* to support development objectives. As such, PSIs engage the private sector, both as (1) a *recipient* of and as (2) a *source* of development finance. PSIs are typically non-concessional, in that they seek financial returns alongside development impact.

share of total commitments (Figure 5), private philanthropic institutions (1.2%) and non-DAC countries (0.1%)[9] — typically non-Western countries — provided climate finance exclusively on concessional terms.

# MAJOR RECIPIENTS OF CLIMATE FINANCE

**In absolute terms, *India* received the largest amount of climate finance** ($72.74 billion), **at more than twice that of *Bangladesh*** ($33.07 billion), **followed by the *Philippines*** ($25.37 billion), ***Indonesia*** ($24.56 billion), and ***Turkiye*** ($23.37 billion) (Figure 8). Characterized by high environmental vulnerability and lack of environmental resilience, ***China*** ($18.66 billion) ranked as the 6th largest recipient of climate finance, though it fared relatively well on the economic vulnerability-resilience dimension.

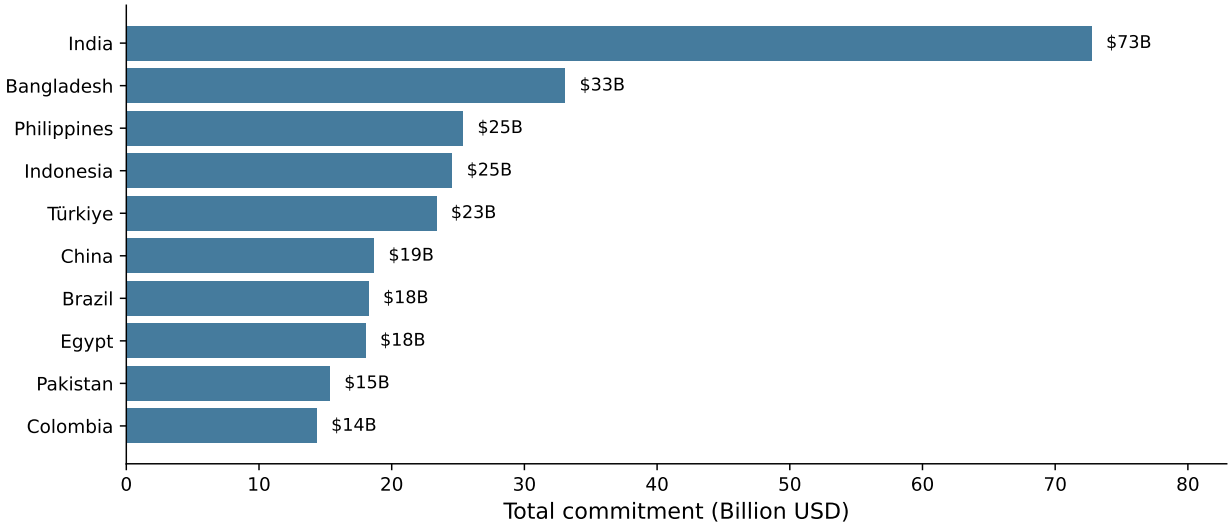


Figure 8: Top 10 Recipients of Climate Finance, by Total Commitment (2015-2023)

**When adjusted for the population, however, Small Island Developing States (SIDS) emerged as the major recipients of climate-related development finance**, exhibiting a very different picture (Figure 9). In fact, all of the top 10 largest recipients were dominated by SIDS: *Tuvalu*, *Nauru*, *Palau*, *Marshall Islands*, *Dominica*, *Tonga*, *Kiribati*, *Vanuatu*, *Micronesia*, *Saint Vincent and the Grenadines*. Among non-SIDS, *Costa Rica*, *Georgia*, *Montenegro*, *Serbia*, and *Albania* recieved the highest per capita allocations, though at levels far below those of the top 10 SIDS recipients.

[9] Non-DAC countries that provided climate-related development finance during 2015-2023 included: *Azerbaijan*, *Croatia*, *Kazakhstan*, *Liechtenstein*, *Monaco*, *Qatar*, *Romania*, and *the UAE*.

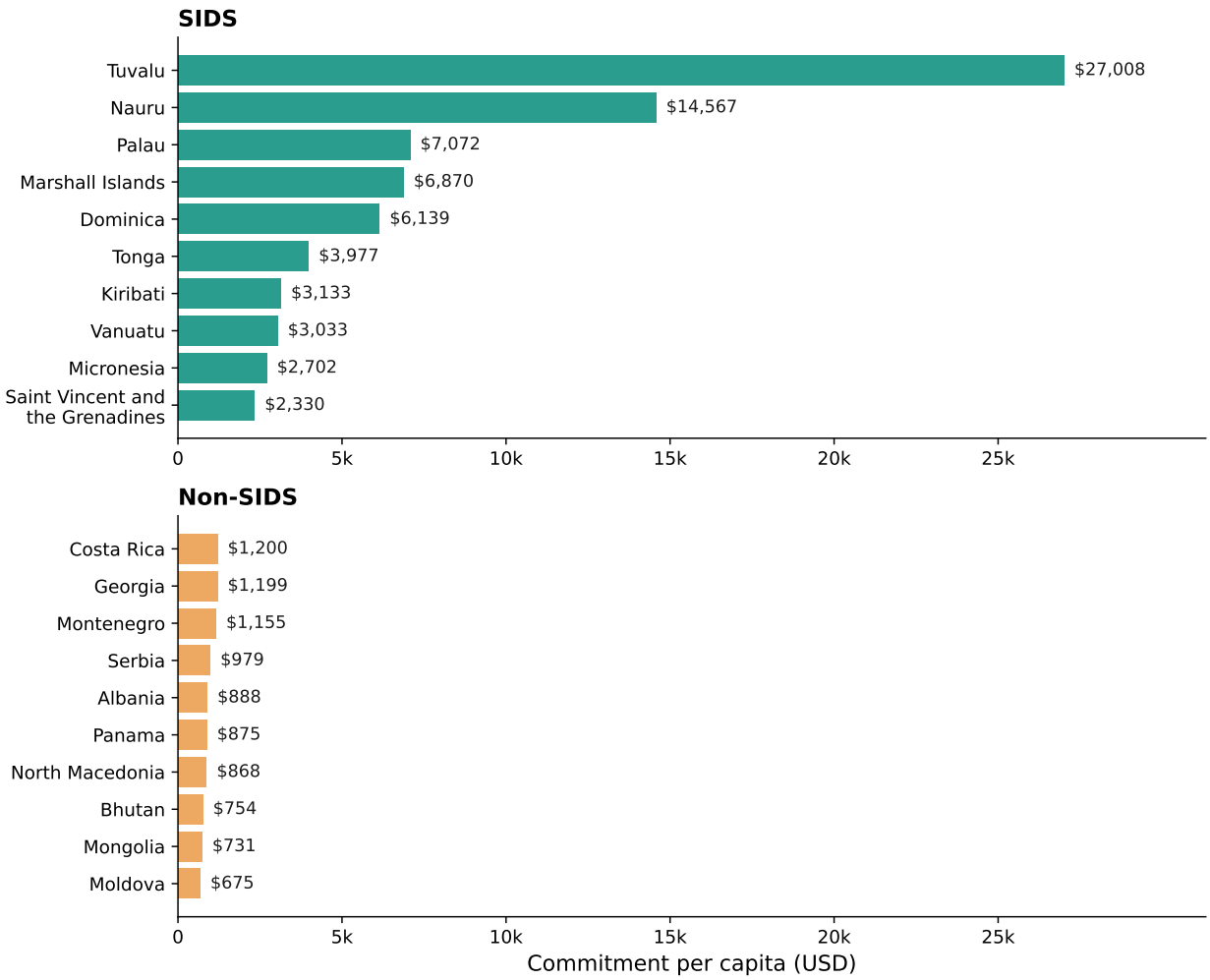


Figure 9: Top 10 Recipients, by Climate Finance per Capita (2015-2023)

These results show that neither measures are perfect: the raw total overrepresents large, populous countries, while per capita values overrepresent small countries in the rankings. Taking note of these methodological constraints as well as the long-tail distribution of the data (the presence of extreme values), subsequent analyses examined $\log(\text{commitment per capita})$ to ensure comparability among countries with vastly different population sizes, while minimizing the influence of outliers (Figure 10).

# RESPONSIVENESS TO DEVELOPING COUNTRIES' NEEDS

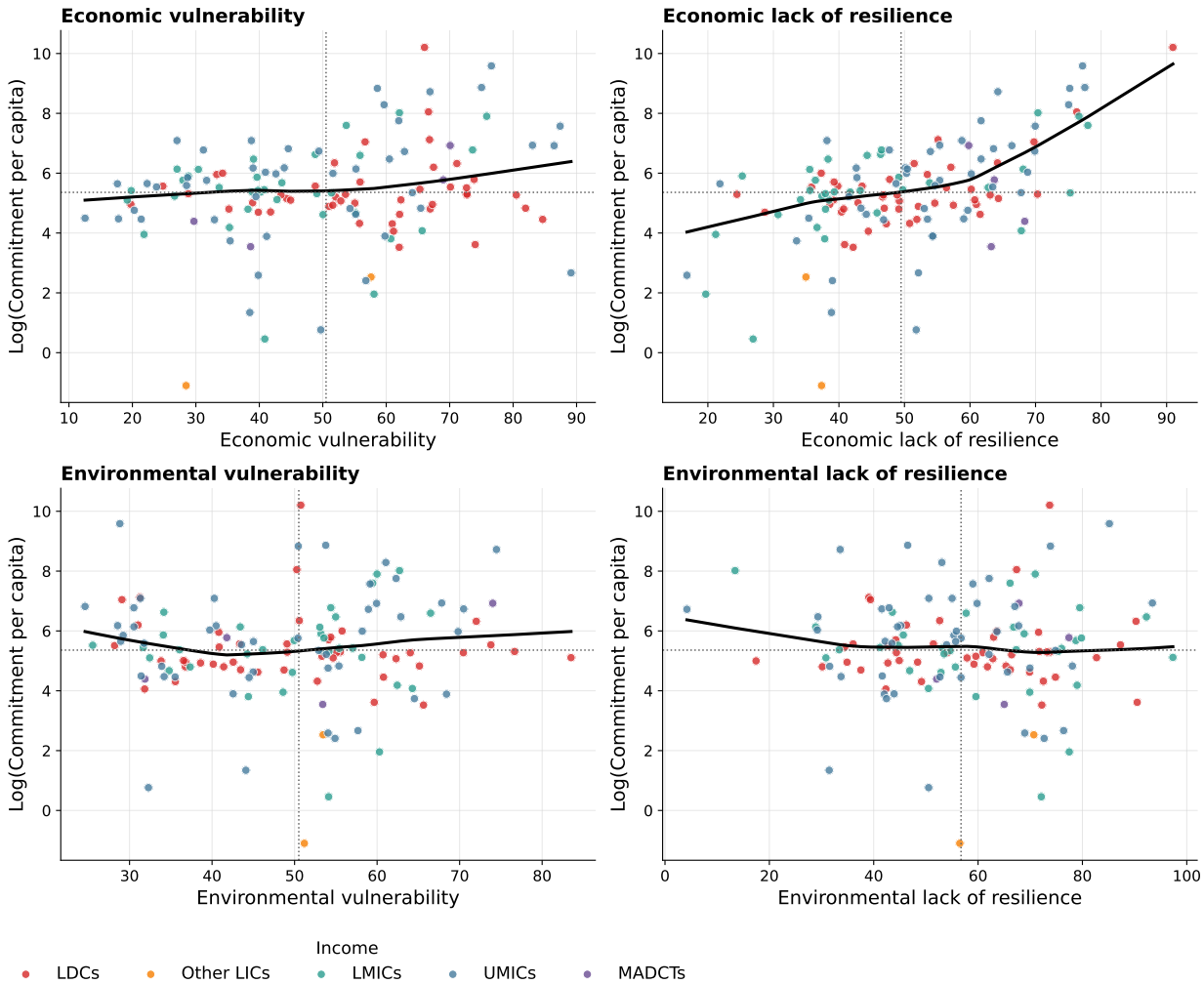


Figure 10: The Relationship between Climate Finance Allocations and Multidimensional Vulnerability Index (MVI)

**Overall, there is mixed evidence that climate finance is provided to countries most in need.**[10] According to the scatterplot, **countries that lack *economic resilience* tend to receive more finance** ($\rho$ = 0.46). **The same is true of countries with high *economic***

[10] The UN Multidimensional Vulnerability Index (MVI) is a comprehensive assessment of a country's well-being. MVI captures two domains: (1) *structural vulnerability* (the risk of a country's sustainable development being hindered by recurrent adverse exogenous shocks); and (2) *structural resilience* (the *inherent* characteristics or *inherited* capacity of a country to minimize, withstand, and recover from such shocks). Each domain is assessed against the three dimensions of sustainable development: (1) *economic*, (2) *environmental*, and (3) *social*. MVI scores range from 0 to 100, with higher values indicating greater vulnerability or lack of resilience. Note that *social* dimensions of MVI were excluded from this analysis, as they have less bearing on the tackling of the triple planetary crisis.

***vulnerability***, although the positive association is much weaker ($\rho$ = 0.18). By contrast, *environmental* dimensions do not bear any relationship with climate finance allocations ($|\rho| < 0.05$). As illustrated by the wide vertical dispersion of the data points along the y-axis, **countries with similar *environmental vulnerability* and *environmental resilience* scores often receive very different amounts of climate finance**. Particularly, *Libya*, *Somalia*, *Eritrea*, *Lesotho*, *Botswana*, *Afghanistan*, *Yemen*, and *Sudan* are underfunded, despite high exposure to both environmental and economic shocks coupled with lack of capacity to withstand such stressors.

Taken together, allocation decisions must be better targeted, for projects to deliver results on-the-ground. Importantly, **climate finance should be more responsive to the *environmental* realities of recipient countries - the very risks it is meant to address.**

# APPENDIX

## PORTFOLIOS OF TOP BILATERAL PROVIDERS, BY REGION AND SECTOR

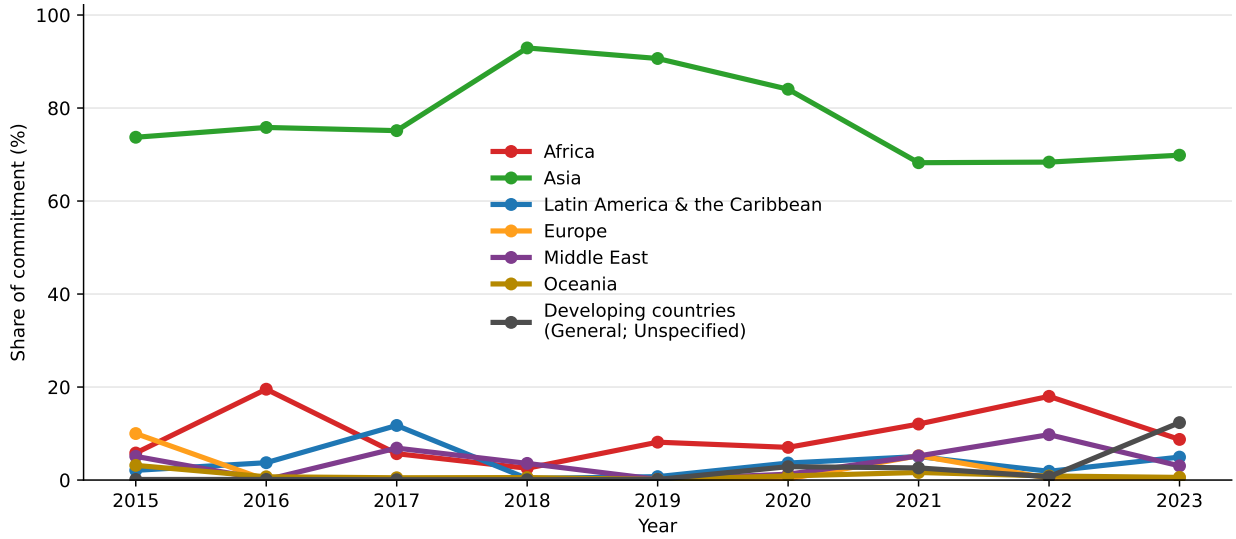


Figure 11: Japan's Climate Finance Allocation, by Region

Table 3: Japan's Top 10 Sectors

| Rank | Sector | Commitment |
|---|---|---|
| 1 | Transport & Storage | $50.50B |
| 2 | Energy | $12.31B |
| 3 | Water Supply & Sanitation | $9.35B |
| 4 | Other Multisector | $5.89B |
| 5 | Agriculture, Forestry, Fishing | $4.67B |
| 6 | Health | $4.53B |
| 7 | Industry, Mining, Construction | $1.47B |
| 8 | Disaster Prevention & Preparedness | $0.95B |
| 9 | General Environment Protection | $0.78B |
| 10 | Emergency Response | $0.50B |

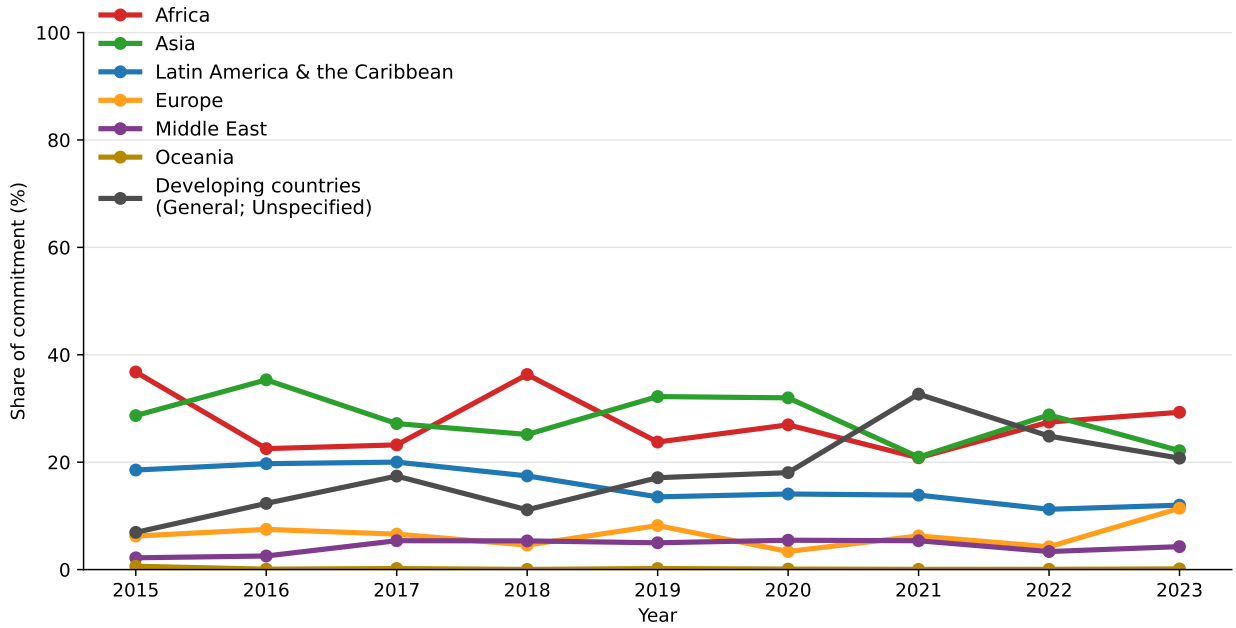


Figure 12: Germany's Climate Finance Allocation, by Region

Table 4: Germany's Top 10 Sectors

| Rank | Sector | Commitment |
|---|---|---|
| 1 | Energy | $22.47B |
| 2 | General Environment Protection | $10.47B |
| 3 | Water Supply & Sanitation | $8.90B |
| 4 | Agriculture, Forestry, Fishing | $7.30B |
| 5 | Other Multisector | $7.09B |
| 6 | Transport & Storage | $5.73B |
| 7 | Banking & Financial Services | $3.47B |
| 8 | Government & Civil Society | $2.75B |
| 9 | Unallocated / Unspecified | $2.47B |
| 10 | Other Social Infrastructure & Services | $1.61B |

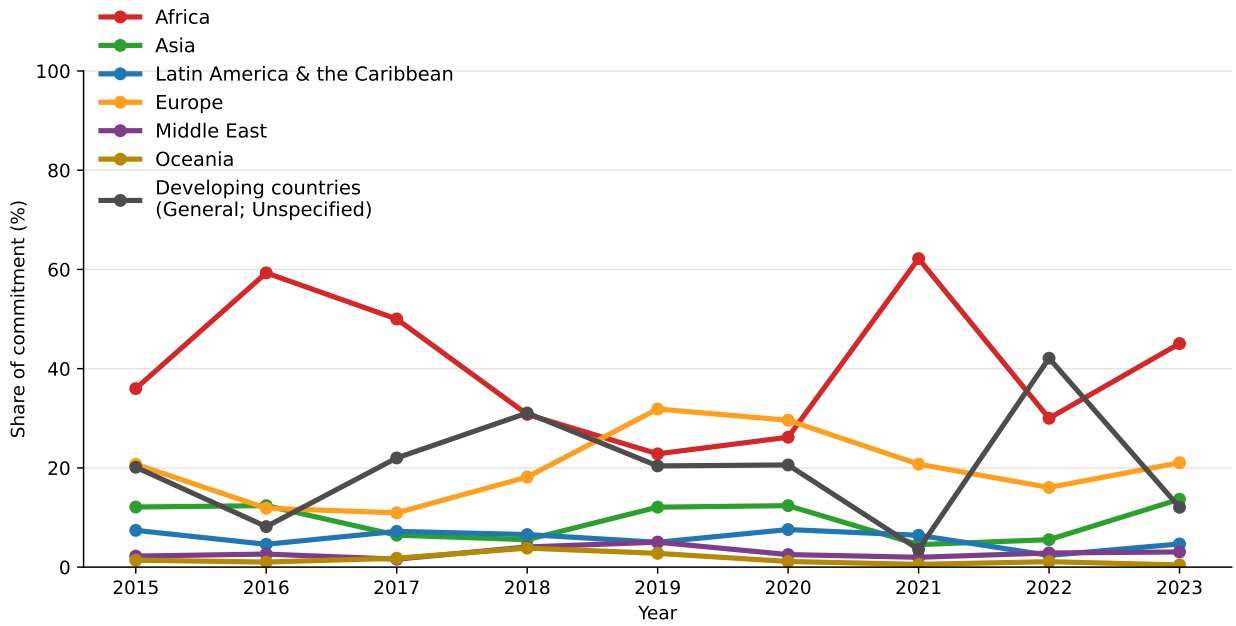


Figure 13: EU's Climate Finance Allocation, by Region

Table 5: EU's Top 10 Sectors

| Rank | Sector | Commitment |
|---|---|---|
| 1 | Agriculture, Forestry, Fishing | $8.65B |
| 2 | Other Multisector | $7.06B |
| 3 | Energy | $6.89B |
| 4 | General Environment Protection | $4.69B |
| 5 | Water Supply & Sanitation | $3.92B |
| 6 | Transport & Storage | $3.88B |
| 7 | Government & Civil Society | $3.20B |
| 8 | Industry, Mining, Construction | $2.01B |
| 9 | Other Social Infrastructure & Services | $1.56B |
| 10 | Health | $1.47B |

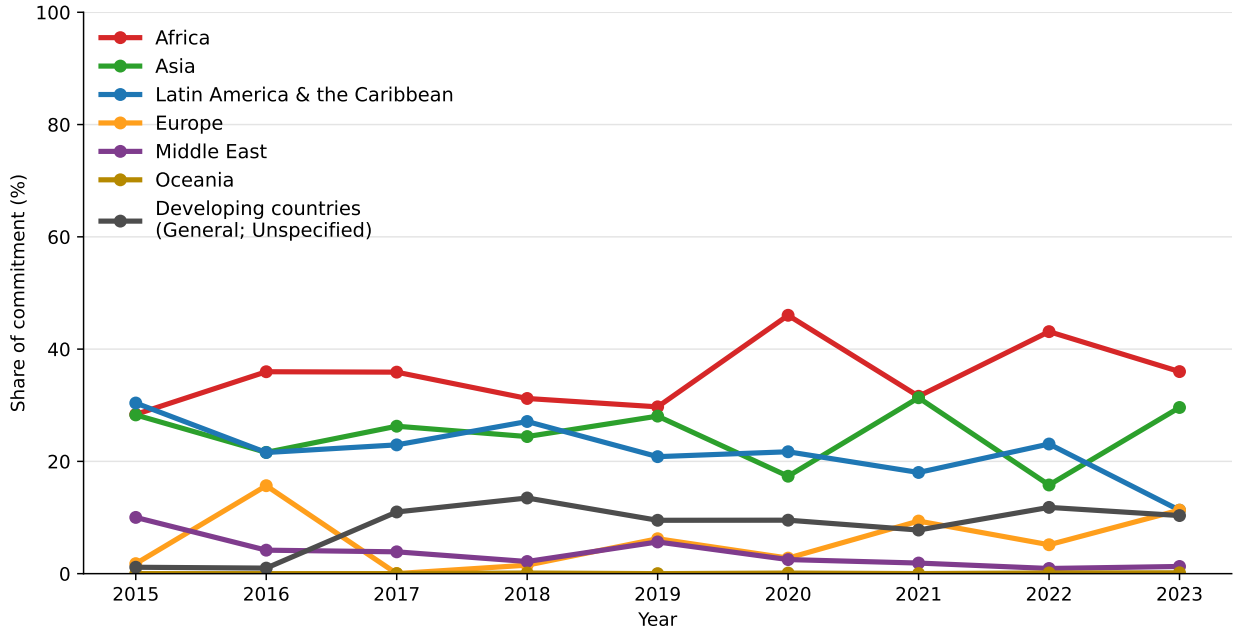


Figure 14: France's Climate Finance Allocation, by Region

Table 6: France's Top 10 Sectors

| Rank | Sector | Commitment |
|---|---|---|
| 1 | Energy | $7.54B |
| 2 | Transport & Storage | $5.94B |
| 3 | Water Supply & Sanitation | $5.89B |
| 4 | Other Multisector | $5.61B |
| 5 | Unallocated / Unspecified | $5.54B |
| 6 | General Environment Protection | $4.62B |
| 7 | Banking & Financial Services | $3.93B |
| 8 | Agriculture, Forestry, Fishing | $3.25B |
| 9 | Government & Civil Society | $2.30B |
| 10 | Other Social Infrastructure & Services | $1.83B |

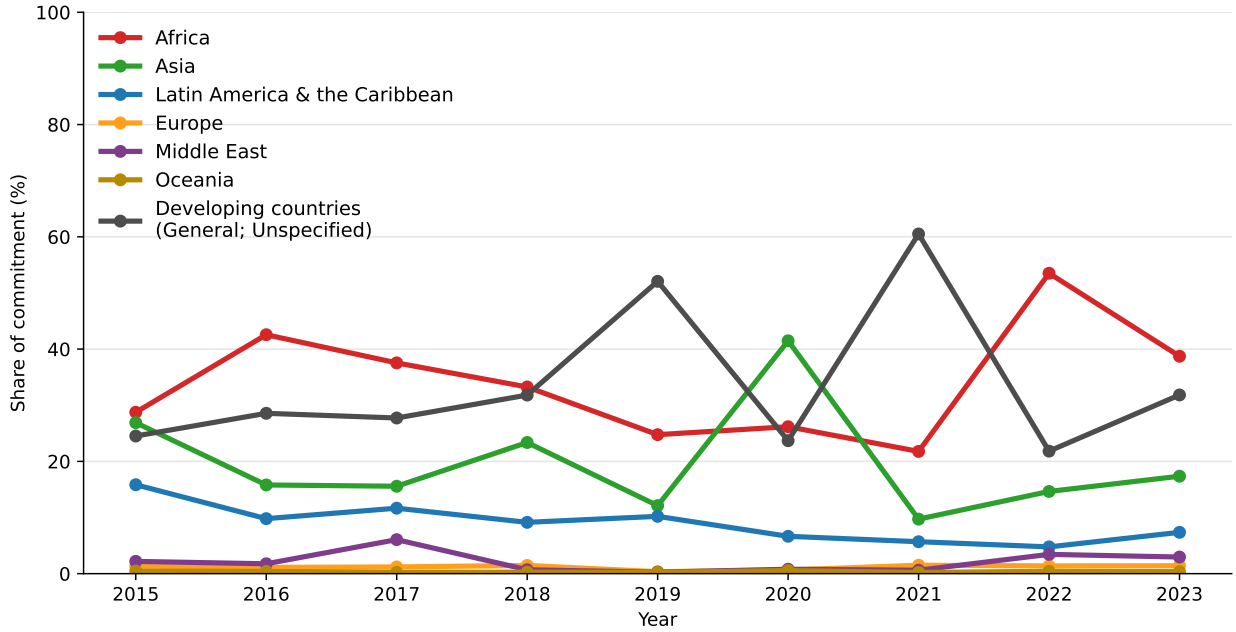


Figure 15: US' Climate Finance Allocation, by Region

Table 7: US' Top 10 Sectors

| Rank | Sector | Commitment |
|---|---|---|
| 1 | Unallocated / Unspecified | $4.26B |
| 2 | Population Policies/Programmes & Reproductive Health | $3.44B |
| 3 | Agriculture, Forestry, Fishing | $3.43B |
| 4 | General Environment Protection | $3.10B |
| 5 | Health | $2.92B |
| 6 | Energy | $1.38B |
| 7 | Business & Other Services | $1.19B |
| 8 | Water Supply & Sanitation | $0.59B |
| 9 | Government & Civil Society | $0.55B |
| 10 | Development Food Assistance | $0.45B |

# SHARE OF CONCESSIONAL VS. NON-CONCESSIONAL CLIMATE FINANCE, BY REGION

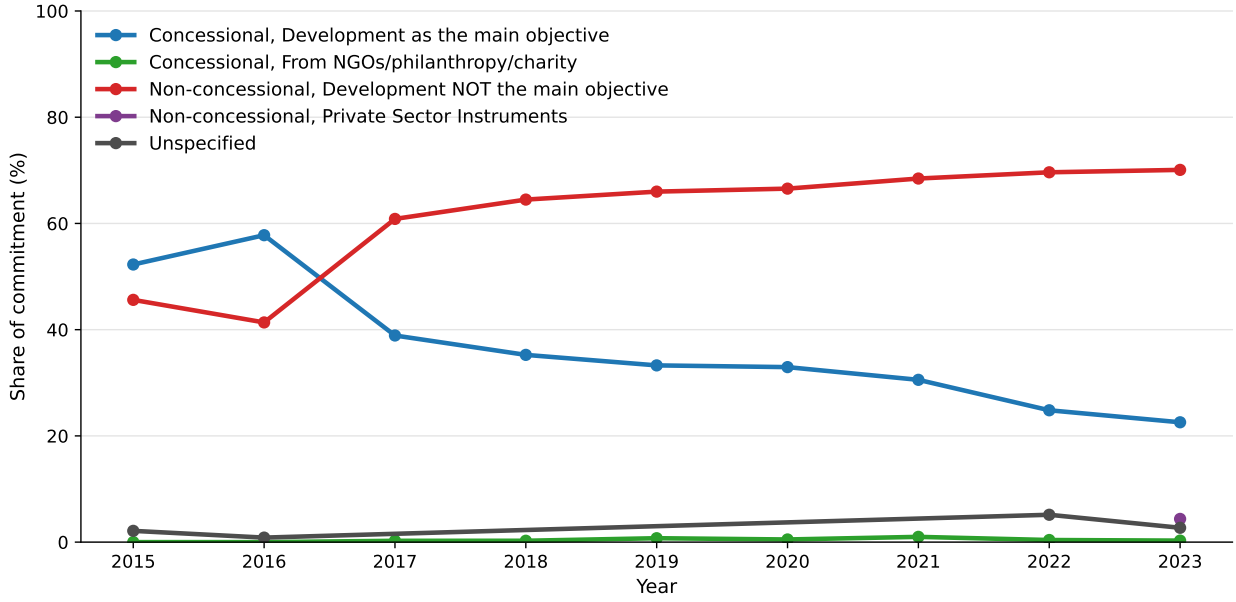


Figure 16: Concessionality Mix in Latin America & the Caribbean

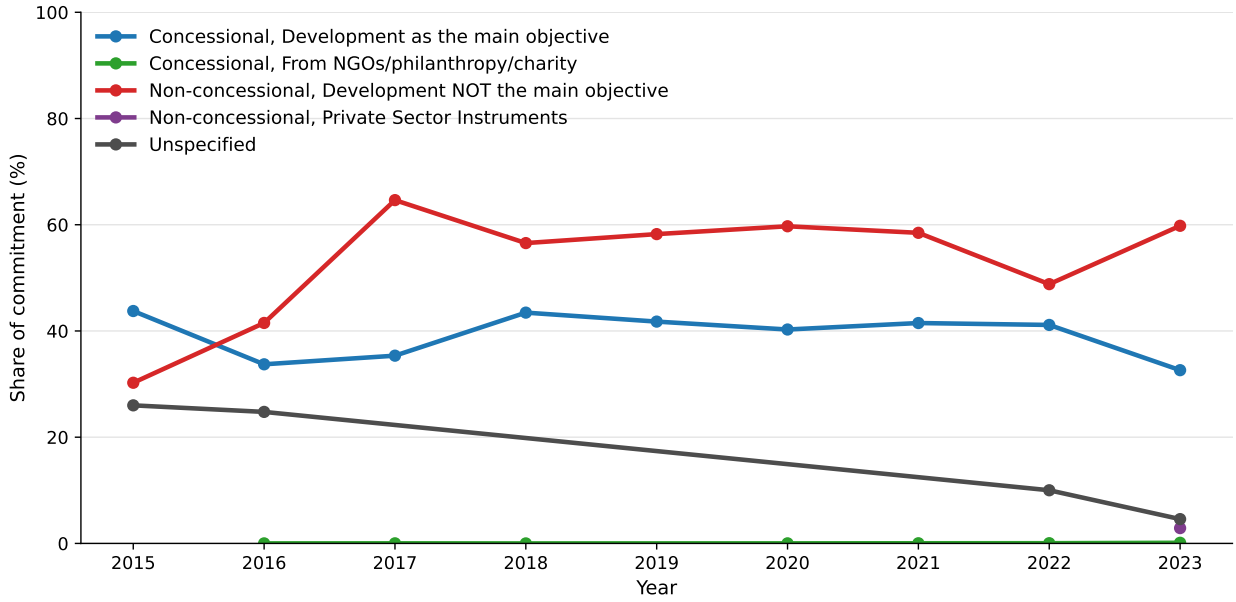


Figure 17: Concessionality Mix in Europe